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An Assessment of an Unsuccessful Restoration Project for Lake Sturgeon Using Three-Dimensional Numerical Modelling

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Abstract

Despite a widespread acknowledgment that river restoration projects sometimes fail due to a poor understanding of geomorphology and hydrology, there are relatively few published case studies reporting failures, particularly for non-salmonid species such as Lake Sturgeon *Acipenser fulvescens*. We used a three-dimensional hydrodynamic model to retroactively assess a restoration project in the 80-m-wide Ouareau River, Quebec which did not meet its objective of providing additional spawning habitat for Lake Sturgeon. Virtual modifications of the bathymetry allowed for the flow field to be simulated with and without instream structures (boulder weirs) constructed in 2007 for four discharges representing flow conditions during spawning. Simulated velocities and flow depths were used to determine the suitability of the site and to assess the impact of the instream structures. Results revealed that instream structures did not meet the expectation of raising water levels and had no significant impact on river velocity. Furthermore, there was sufficient good quality habitat within the study area before restoration, and artificial spawning sites were placed in locations with nonoptimal velocities for spawning Lake Sturgeon. A comparison with a successful Lake Sturgeon restoration project in the St. Clair–Detroit River system revealed marked differences in restoration strategies, which likely explain the different outcomes of these two projects. These results point to the need for (1) adaptive management protocols that include an iterative decision-making process to allow for adjustments in hypotheses and strategies to improve the management process, (2) multidisciplinary input, including hydrogeomorphology, and (3) a more systematic use of two-dimensional or three-dimensional numerical models prior to the implementation of instream structures in river restoration projects.

River restoration projects are often undertaken to counteract biodiversity loss due to habitat degradation (Palmer et al. 2005). In freshwater environments, an exponential growth of restoration projects was observed in the 1990s, and the most recent estimates showed costs exceed a billion dollars annually in the United States (Bernhardt et al.

2005; Lapointe et al. 2013). These projects often involve the manipulation of river form and function through the introduction of instream structures, which increase the cost of restoration projects but are rarely assessed for performance (Bernhardt et al. 2005; O'Donnell and Galat 2007; Whiteway et al. 2010).

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It is widely recognized that river restoration projects sometimes fail due to a poor understanding of the geomorphology or hydrology of the river (Frissell and Nawa 1992; Kondolf 1998; Downs and Kondolf 2002; Woolsey et al. 2007). However, too often the failed projects are left undocumented (Paillex et al. 2017), so that important lessons cannot be learned from these past projects (Bash and Ryan 2002; Downs and Kondolf 2002; Woolsey et al. 2007).

The lack of critical assessment of restoration projects is even more acute for nonsalmonid species, such as the Lake Sturgeon *Acipenser fulvescens*, which is classified as either endangered, threatened, or of special concern at the state level in the United States and the federal level in Canada. In many cases, spawning habitat is considered as a limiting factor for population growth (COSEWIC 2006; Lyttle 2008; Bennion and Manny 2014; Krieger and Diana 2017). To date, 17 projects have attempted to restore or create suitable spawning habitat for Lake Sturgeon throughout North America (Baril et al. 2018). These projects address two main issues: (1) to compensate for lost habitat from anthropogenic disturbances (Johnson et al. 2006; Manny et al. 2015); or (2) to expand existing habitat to address an insufficient quantity of suitable habitat at the site (Dumont et al. 2011).

A Lake Sturgeon rehabilitation project was initiated in 2007 on the Ouareau River, Quebec, a relatively large, gravel-bed river (width of around 80 m). The Ouareau River site was deemed to have experienced a loss of quantity and function of traditional spawning habitat due to a landslide event in March 1990. The need for active restoration of the Ouareau River was mainly driven by the severe decline in the Lake Sturgeon population size. This was believed to be due to a loss of adequate spawning habitat, and the following three hypotheses were proposed: (1) the lack of suitable habitat resulted in egg densities that were too high, likely causing high embryonic mortality rates; (2) headward erosion caused degradation of the spawning reach; and (3) water level fluctuations, related to dam activity upstream, resulted in unsuitable conditions for Lake Sturgeon spawning (Beaulieu and Bélanger 2005). The project installed three artificial spawning reefs in areas that were characterized as having lost suitable habitat for spawning Lake Sturgeon and constructed two arch-shaped boulder weirs to increase water levels and resist headward erosion (Beaulieu and Bélanger 2005). However, egg and larval monitoring surveys after construction revealed that the artificial spawning beds were not being used by Lake Sturgeon, and the downstream weir was rapidly degrading. Although boulder weirs can have positive effects on salmonids in relatively small rivers (<20 m wide: Roni et al. 2008), their use in large rivers such as the Ouareau River, or for Lake Sturgeon

restorations in general, is unprecedented (Sear et al. 2010; Baril et al. 2018).

The failure of the Ouareau River project to achieve its restoration goal highlights the need to understand the dynamics of a river reach before undertaking any restoration measures. Two-dimensional (2D) models have been used with success in ecohydraulic assessment and river restoration design to represent the spatial variability of velocity around instream structures (Pasternack et al. 2004; Brown and Pasternack 2009; Pasternack 2011; Pasternack and Brown 2013). However, three-dimensional (3D) numerical models add useful information on the vertical variability in velocity and are particularly useful to represent detailed flow patterns around submerged instream structures (Biron et al. 2012; Khosronejad et al. 2015). One main advantage of 2D or 3D models is to test different scenarios using virtual modification of the bathymetry (e.g., Biron et al. 2012; Jackson et al. 2015), for example, by running tests with and without instream structures. The detailed flow field data predicted by 2D or 3D models can then be used to determine hydraulic metrics and, in combination with a habitat suitability index (HSI), to assess instream fish habitat (e.g., Crowder and Diplas 2006; Kozarek et al. 2010; Wang et al. 2017).

The objective of this study was to use a 3D numerical modelling approach to provide insights on an unsuccessful restoration project for Lake Sturgeon in the Ouareau River. More specifically, this research (1) used virtual modifications of the river bathymetry to compare the flow field for discharges consistent with spawning season with and without weirs and (2) computed the suitability of the Ouareau River for Lake Sturgeon spawning for scenarios with and without weirs. This was accomplished by using an HSI developed for Lake Sturgeon spawning (Baril et al. 2018) to conduct a fuzzy multicriteria analysis within a GIS.

METHODS

Study area.—The Ouareau River, a tributary of the L'Assomption River, drains into the St. Lawrence River, Quebec, (Figure 1) where the Ouareau Lake Sturgeon population spends the majority of its life cycle (Fortin et al. 1992). The only known spawning site on the Ouareau River is located 2.6 km downstream from a dam, constructed in 1918, in Crabtree (Figure 1). At this site, which has an area of 6.25 ha, the construction of two weirs and three artificial spawning reefs was completed in February 2007 (Figure 2) to augment three natural spawning reefs (Figure 3). Bankfull discharge is 151 m³/s, and the annual average discharge is 32.5 m³/s. The mean width of the river is approximately 80 m during average conditions until it splits into two channels flowing around an island composed of deposited alluvial material. The site

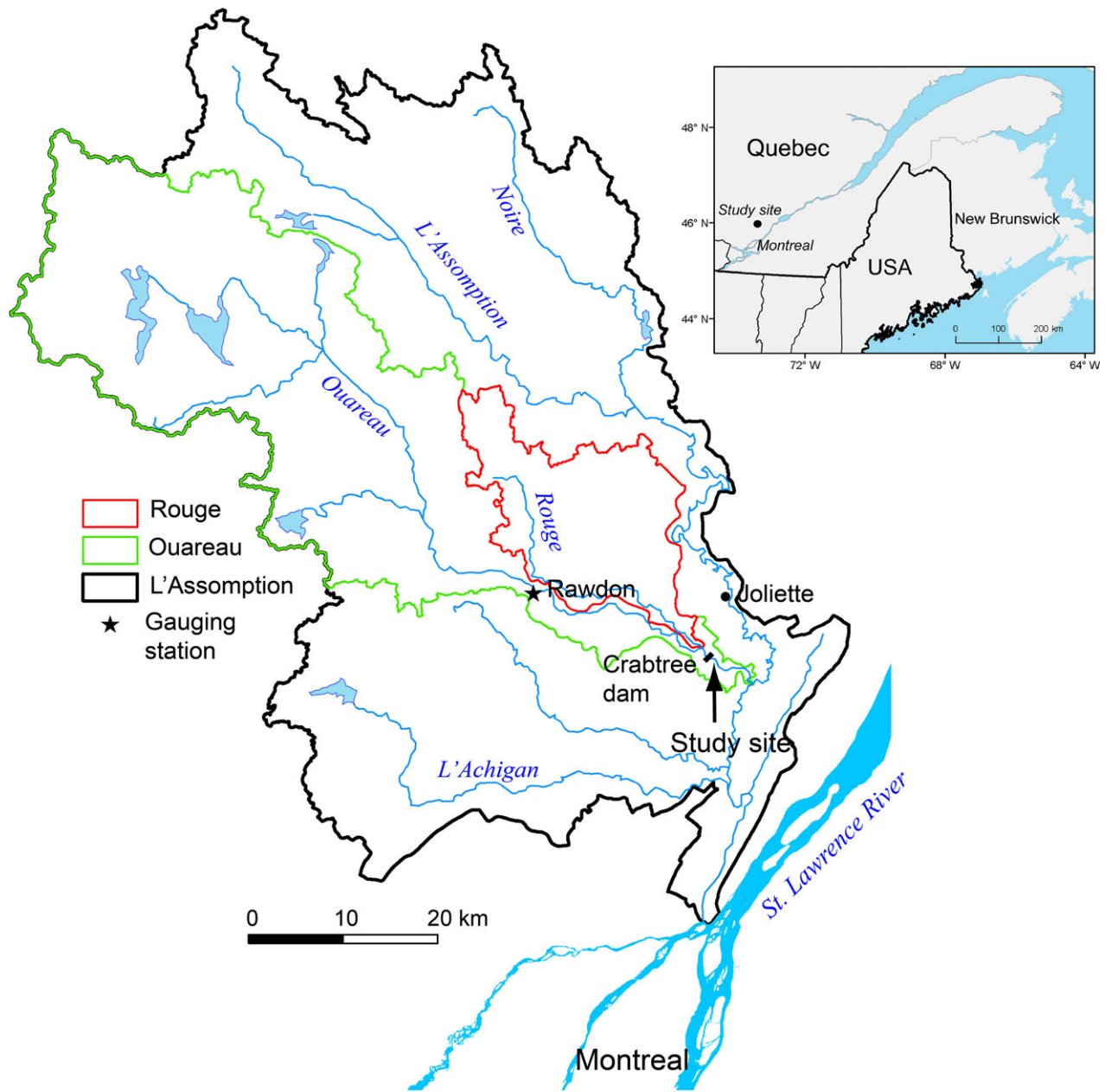


FIGURE 1. Watershed limits of the Ouareau River, Quebec, a tributary of the L'Assomption River, also indicating the Rouge River subwatershed and the Lake Sturgeon spawning site restoration project downstream from the dam in Crabtree. The gauging station in Rawdon (Centre d'Expertise Hydrique du Québec [CEHQ], 052212) is located upstream from the confluence with the Rouge River (black star). [Color figure can be viewed at afs.journals.org.]

has widely variable substrates, ranging from fine clay to boulders, which are generally located in distinct zones (Figure 3).

Field measurements.—A bathymetric survey of the Ouareau River was conducted in 2015 within an area beginning 100 m upstream from the first weir and ending at the natural and artificial traditional spawning sites (Figure 3). These two sites were not included in the detailed analysis described below because they were downstream

from the area affected by the weirs. The bathymetric data were collected with a Spectra Precision SP80 differential GPS (DGPS, precision of ± 1.5 cm). A total of 2,534 measurements were taken along 118 transects, at intervals of approximately 7 m, with additional data taken whenever an abrupt change in bed topography was observed. When adequate satellite reception was not available (usually near the banks), a Zipline Pro (precision of ± 2 cm) was used to measure bed elevation. Wolman pebble counts

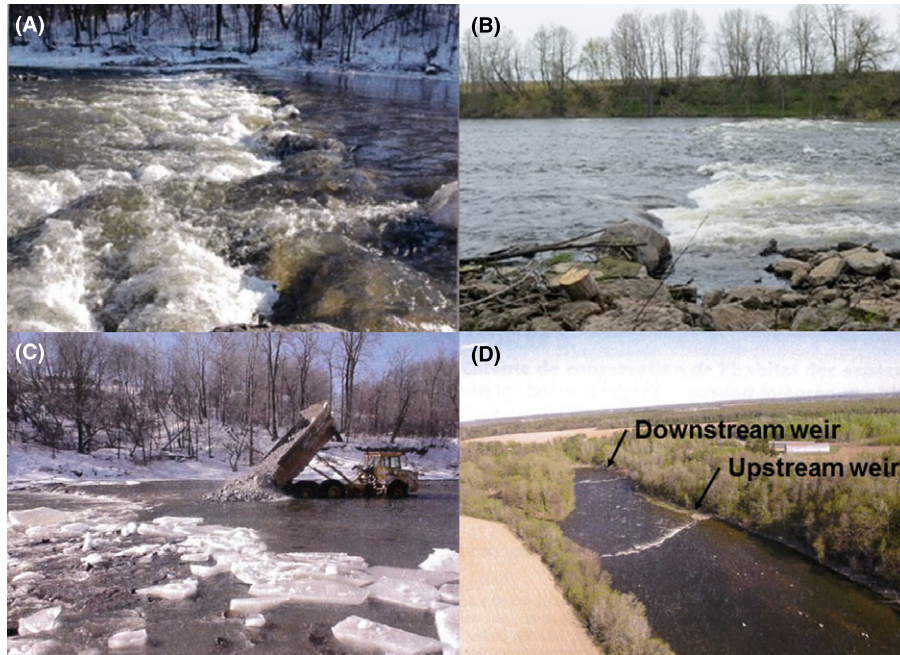


FIGURE 2. (A) Upstream boulder weir immediately following the Lake Sturgeon restoration project in 2007; (B) upstream weir in May 2016 at a discharge of $86 \text{ m}^3/\text{s}$; (C) Installation of the upstream artificial spawning site (upstream from weir in panel A); (D) aerial view of the two constructed weirs in 2007. [Color figure can be viewed at afs-journals.org.]

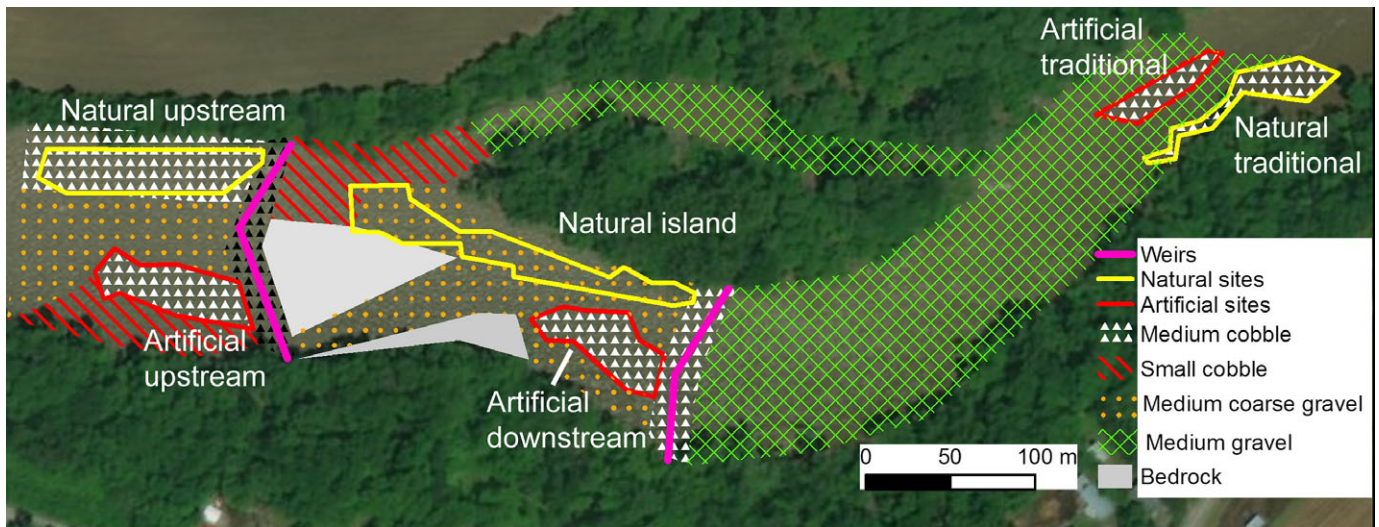


FIGURE 3. Ouareau River indicating artificial (red) and natural (yellow) Lake Sturgeon spawning sites and substrate size, classified using the Wentworth scale (Wentworth 1922). [Color figure can be viewed at afs-journals.org.]

(Wolman 1954) were conducted at 17 locations with distinct substrate sizes (Figure 3) to determine particle size according to the modified Wentworth scale (Wentworth 1922). A pressure transducer (Solinst Levelogger) was installed for 9 weeks from August to October 2015 to determine water levels corresponding to various discharges. Because of the presence of a major tributary, the

Rouge River, between the study area and the gauging station (Figure 1), the site discharge was multiplied by a factor of 1.26 based on the catchment area of the Rouge River (Emerson et al. 2005).

Numerical modelling.—Bathymetric data were used to create a numerical mesh using triangular interpolation in the Delft3D hydrodynamic modelling software (Deltares

2014). The model predicts depth and velocity at each of the 27,435 cells covering the study area (5,487 horizontal cells by five vertical layers, with an average horizontal cell area of 11.4 m² and cell height of 0.13 m). The grid was compartmentalized into the 17 substrate zones, each of which were assigned a Manning's *n* roughness coefficient corresponding to the results from the Wolman pebble count (Kondolf and Piégay 2016). Substrate sizes ranged from a median diameter (D50) of 20 to 109 mm, and the corresponding Manning's *n* ranged between 0.025 and 0.037. Upstream and downstream boundary conditions were discharge and water level, respectively. The model was calibrated using water surface elevation measured with the DGPS for a summer low-flow condition (8.5 m³/s). The model was then validated by comparing predicted and measured (DGPS) water surface elevation measurements at high flow (176 m³/s). The model showed excellent agreement (root mean square error = 0.021 m) relative to other validation examples using water surface elevation (Hodge et al. 2013; Matte et al. 2017).

The discharge values used for the numerical simulations were based on flow conditions occurring between mid-April and early May in the Ouareau River, corresponding to both the observed Lake Sturgeon spawning behavior at this site and 11–15 d before successful larval Lake Sturgeon surveys downstream from the spawning area. This timeline was selected as it takes 8–10 d after fertilization for eggs to hatch, and the larvae spend an additional 3–5 d within the substrate before migrating downstream (Auer and Baker 2002). Using these criteria, four discharges were chosen for Delft3D simulations to represent the range of spawning conditions at the Ouareau River site: 38, 55, 69, and 87 m³/s.

The numerical grid of the Ouareau River bed was virtually modified to represent the preresoration scenario with no weirs by interpolating bed elevation for zones located 5 m upstream and downstream from weirs. The same four discharge conditions were used to run these no-weir Delft3D simulations. The predicted longitudinal water surface profiles for conditions with and without weirs were examined to determine the effect of the weirs

on water levels for the lowest spawning discharge (38 m³/s) when the weirs should have the most significant effect on water levels (Bariteau and Duhamel 2008).

GIS and fuzzy analysis.—Results from the hydrodynamic model were exported to a GIS (ArcGIS 10.4) where average velocities and depths within the spawning sites were compared with an HSI for Lake Sturgeon spawning (Baril et al. 2018) for each spawning zone, both with and without weirs. Water temperature was not included in the HSI fuzzy analysis described below because it does not vary spatially within a reach, and each spring (usually in May) suitable temperatures will occur in the Ouareau River.

A fuzzy evaluation was used to determine the final suitability score. This is a form of evaluation in which the truth values of variables (here, suitable velocity, depth, or substrate) may be any real number between 0 and 1, allowing for a continuous gradient of “membership” to the suitable sets (Malczewski 1999). The fuzzy membership function also uses the minimum thresholds of suitability, determined for each factor by Baril et al. (2018), corresponding to 0.7, 0.8, and 0.9, respectively, for velocity, substrate, and depth (Table 1; Figure 4).

The HSI curve of Baril et al. (2018) for velocity shows that suitability increases rapidly until reaching a peak at 0.6 m/s; it then slowly declines until reaching a value of 0 at 2.6 m/s (Table 1). This curve was converted into a fuzzy membership function (Figure 4A) showing a maximum fuzzy score of 1 for velocities near 0.6 m/s and high fuzzy scores (>0.7) for values down to 0.42 m/s and up to 1.35 m/s (Table 1; Figure 4). For depth, the fuzzy function depicted in Figure 4B followed the HSI curve in Figure 5 of Baril et al. (2018) for small rivers, where the annual discharge is less than 100 m³/s. Depth suitability increased from 0.26 m to a peak between 0.55 and 0.85 m, before slowly declining until 2.05 m, at which point suitability rapidly dropped until becoming unsuitable at 3.15 m (Table 1; Figure 4B). For substrate, cobbles were assigned the maximum fuzzy value of 1, pebbles, gravel, and boulders were assigned high fuzzy scores of 0.84, 0.77, and 0.57, respectively, and particles

TABLE 1. Values for the fuzzy analysis of HIS for the three criteria of velocity, depth, and substrate. High fuzzy values are considered as good habitat, exceeding the high fuzzy threshold (from Baril et al. 2018).

Criteria	Minimum–maximum	High fuzzy score threshold	Values exceeding high fuzzy threshold	Peak values (fuzzy score = 1)
Velocity	0.1–2.6 m/s	0.7	0.42–1.35 m/s	0.6 m/s
Depth	0.03–3.15 m	0.9	0.40–2.05 m	0.55–0.85 m
Substrate	≤2 mm (sand or smaller)–4,096 mm (boulder)	0.8	2.1–4,096 mm (gravel to boulder)	64–256 mm (cobble)

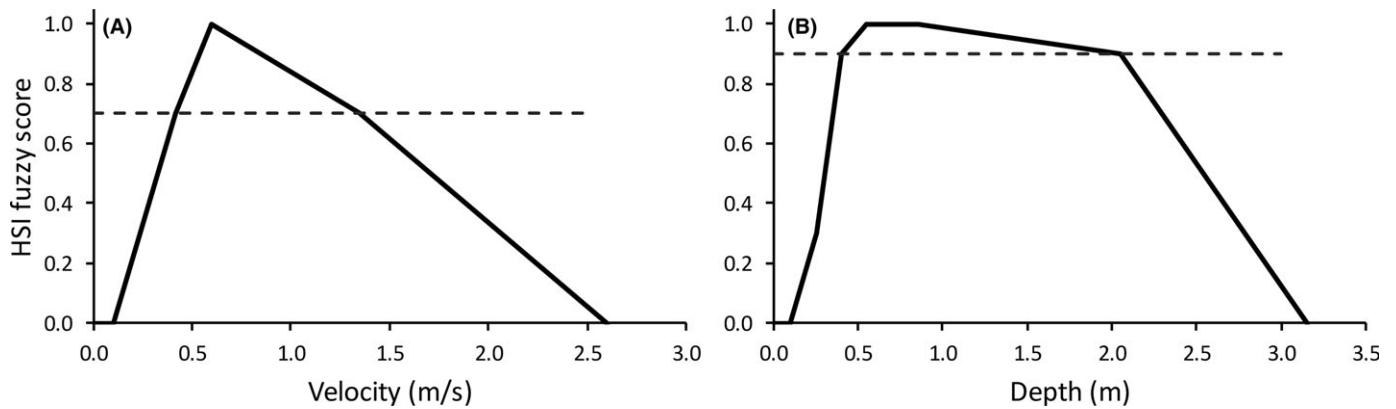


FIGURE 4. Habitat suitability index curves for (A) velocity and (B) depth, adopted from Baril et al. (2018). The dashed gray line represents the threshold for "excellent" habitat (HSI = 0.7 for velocity and 0.9 for depth).

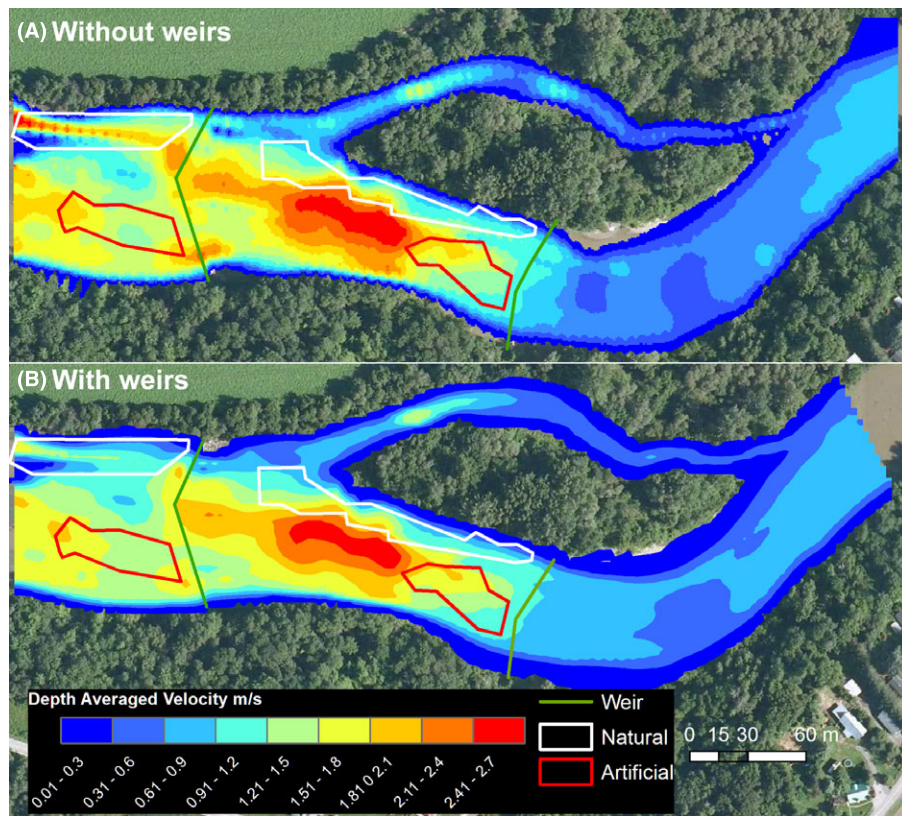


FIGURE 5. Velocity for an average Lake Sturgeon spawning flow condition of $55 \text{ m}^3/\text{s}$ (A) with weirs and (B) without weirs. [Color figure can be viewed at afsjournals.org.]

finer than gravel and bedrock were considered unsuitable for Lake Sturgeon spawning.

To assess habitat suitability within the study reach, a fuzzy multicriteria evaluation using weighted linear combination (WLC) was conducted using GIS with the three factors of velocity, depth, and substrate converted to a fuzzy score. Each factor was assigned a weight based on published results, which indicates that the primary drivers

of Lake Sturgeon spawning site selection are velocity and substrate size, depth has a less important role (Baril et al. 2018). The resulting suitability map thus exhibited a continuous scale ranging from 0 (least suitable) to 1 (most suitable) for each pixel. A sensitivity analysis was also conducted comparing the mean suitability of all spawning beds for the four spawning discharges using four slightly different series of weights ranging between 0.40 and 0.50

for both velocity and substrate, and between 0.10 and 0.20 for depth. This sensitivity analysis revealed that the rank of suitability for the four spawning sites did not change, i.e., the sites with the highest or lowest ranking remained the same with small changes in weight. Thus, the results were not very sensitive to the choice of weights for velocity, substrate, and depth. The weights that provided the best match with known spawning sites were chosen: 0.40, 0.50, and 0.10 for velocity, substrate, and depth, respectively.

Results for the fuzzy WLC habitat suitability model were analyzed for each spawning discharge by first determining the mean HSI score within each of the four spawning beds. Secondly, through a process called “defuzzification” (Malczewski 1999), a weighted mean was used to establish a minimum threshold total suitability based on the minimum thresholds determined for each factor (0.7, 0.8, and 0.9, respectively, for velocity, substrate, and depth (Table 1). To compute the total suitability threshold, the minimum threshold value was calculated as the sum of the products of the weighting factors and each variable’s threshold value. Suitable spawning areas were therefore determined to be cells that exceeded a total suitability score of 0.77, i.e., $(0.4 \times 0.7) + (0.5 \times 0.8) + (0.1 \times 0.9) = 0.77$.

Because fuzzy models incorporate the uncertainty of boundaries set for habitat parameters, it is considered an appropriate technique in species distribution models (Mouton et al. 2011) as it provides more flexibility than the simple Boolean overlay approach, where suitability for each variable (velocity, depth, or substrate) is a simple binary true or false output. The fuzzy approach uses continuous variables and allows a criterion with low suitability to be compensated by the high suitability of another factor (Pechanek and Machar 2013). Using this weighted, fuzzy approach is therefore an improvement over the traditional unweighted HSI approach (e.g., Wang et al. 2017).

RESULTS

Flow velocity during spawning conditions did not vary much with and without weirs (Figure 5), and depth-averaged values were above 2 m/s in some of the central sections of the river. For all discharges, the differences in velocity (Figure 6) and depth (Figure 7) between the weir and no-weir scenarios were minimal, and thus had no effect on the total fuzzy suitability score since, for a given discharge, the weir and no-weir simulations were always either within or outside the high suitability habitat range (green zones in Figures 6, 7). The natural upstream site showed the largest change in velocity and depth where weirs were installed, largely as a result of the removal of rip-rap at the edge of weir 1 during grid creation for

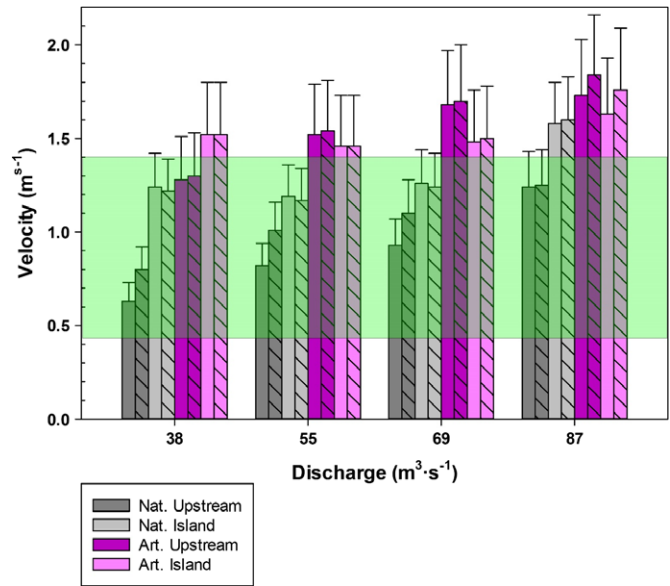


FIGURE 6. Depth-averaged velocity at the Ouareau River Lake Sturgeon spawning sites with (solid color) and without (hatched) weirs for the four representative spawning discharges. The error bars represent the 95% CI. The mean range of depth-averaged velocity from the habitat suitability index is indicated in green fill. [Color figure can be viewed at afs-journals.org.]

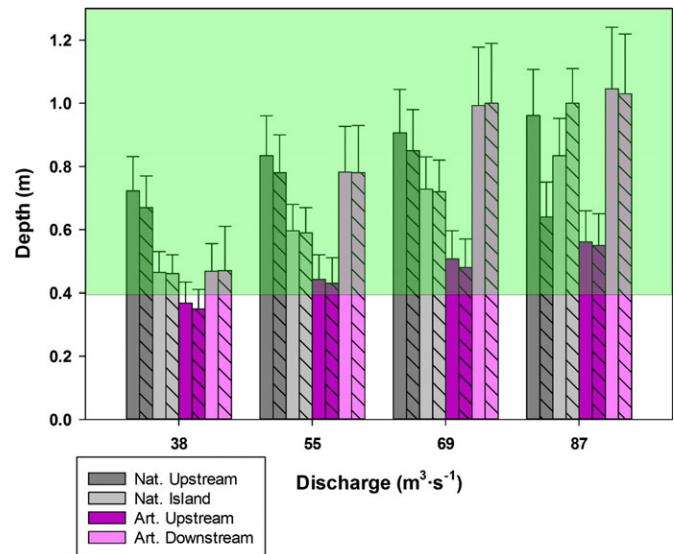


FIGURE 7. Depth at the Ouareau River Lake Sturgeon spawning sites for conditions with (solid color) and without (hatched) weirs for the four representative spawning discharges. The error bars represent the 95% CI. The mean range of depth from the habitat suitability index is indicated with the green fill. [Color figure can be viewed at afs-journals.org.]

simulations without weirs. The cumulative mean difference in velocity at spawning sites across all discharges was only 0.05 m/s (95% CI = 0.01), and the cumulative mean difference in depth was only 0.02 m (95% CI = 0.02). Based on

the habitat suitability curves in Baril et al. (2018), these changes in velocity and depth caused by the weirs would have negligible effects on spawning Lake Sturgeon.

The lack of suitable habitat was believed to be the cause of the Lake Sturgeon population decline at the Ouareau River. However, the combined 3D hydrodynamic modelling and fuzzy analysis revealed there were many suitable spawning zones for Lake Sturgeon in the study

area for the entire range of tested discharges (Figure 8). Because the weight of velocity was greater than for depth, and substrate does not vary with flow conditions, velocity primarily determined how HSI for Lake Sturgeon varied with discharge (Figure 8). Suitable spawning-zone areas ranged from 9,981 m² for a discharge of 38 m³/s to 5,688 m² for a discharge of 87 m³/s. Since the differences in velocity between the weir and no-weir conditions were

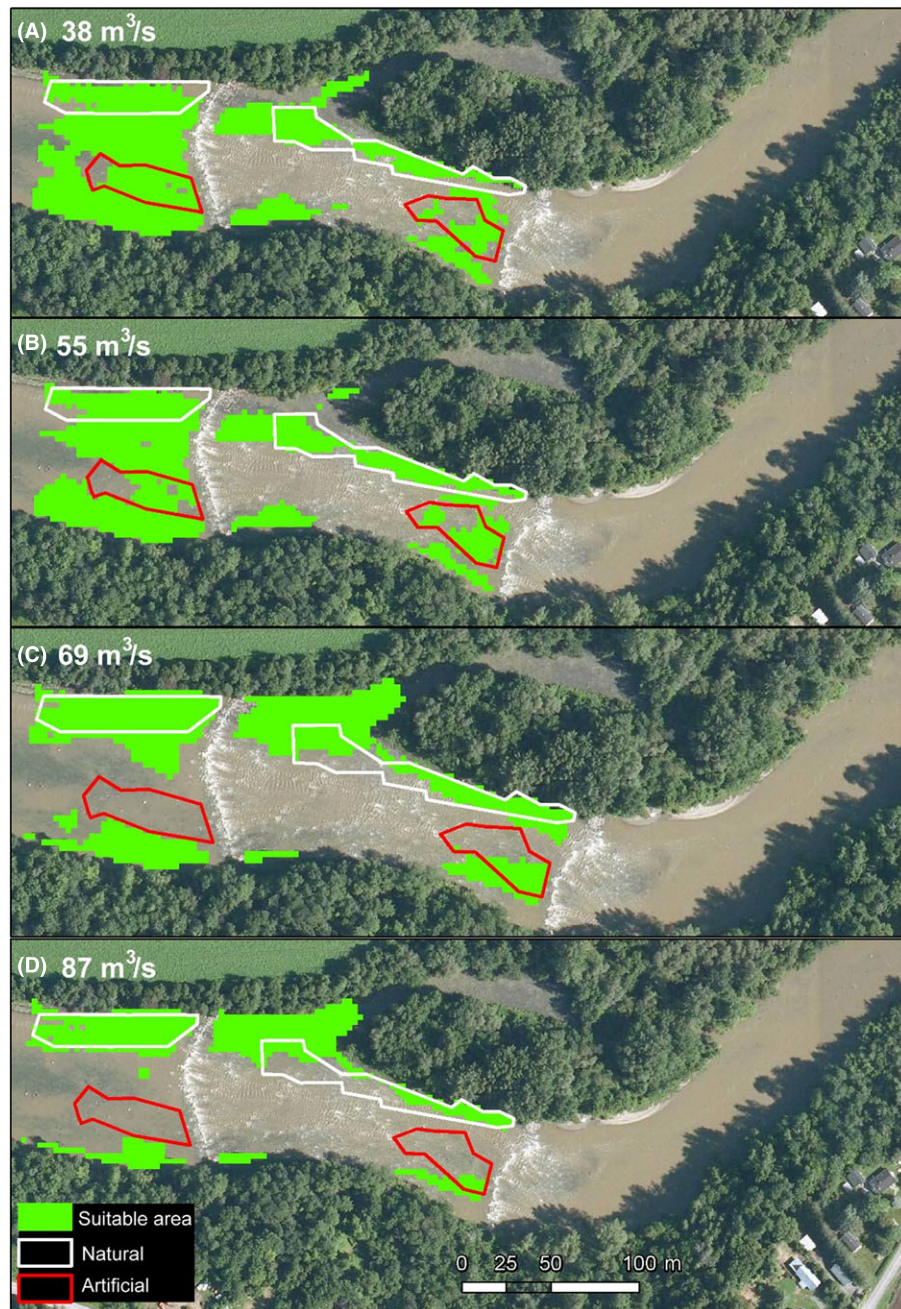


FIGURE 8. Results for the spawning fuzzy habitat suitability model for postrestoration (with weirs) conditions for Lake Sturgeon at discharges of (A) 38 m³/s, (B) 55 m³/s, (C) 69 m³/s, and (D) 87 m³/s. [Color figure can be viewed at afs.journals.org.]

very small (Figure 6), the same suitable spawning zones as highlighted in Figure 8 were present in the no-weir simulations.

Interestingly, suitable habitat in postrestoration conditions was predominantly located outside of the documented natural and artificial spawning sites for all discharges (Figure 9). The artificial sites contributed a

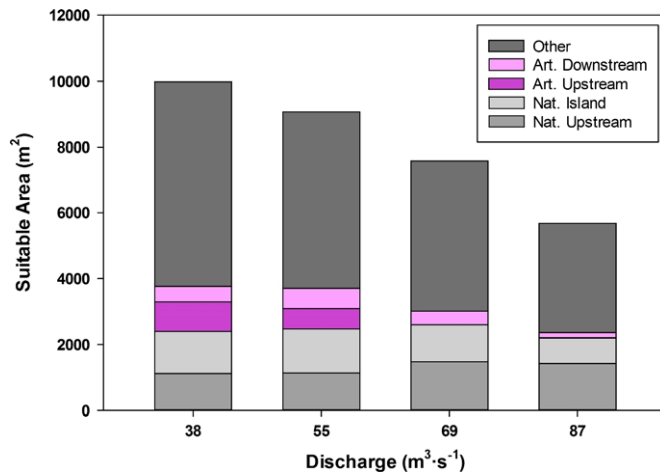


FIGURE 9. The postrestoration amount of suitable spawning area (m²) for Lake Sturgeon inside and outside known spawning sites within the Ouareau River study area for the four representative spawning discharges. [Color figure can be viewed at afs-journals.org.]

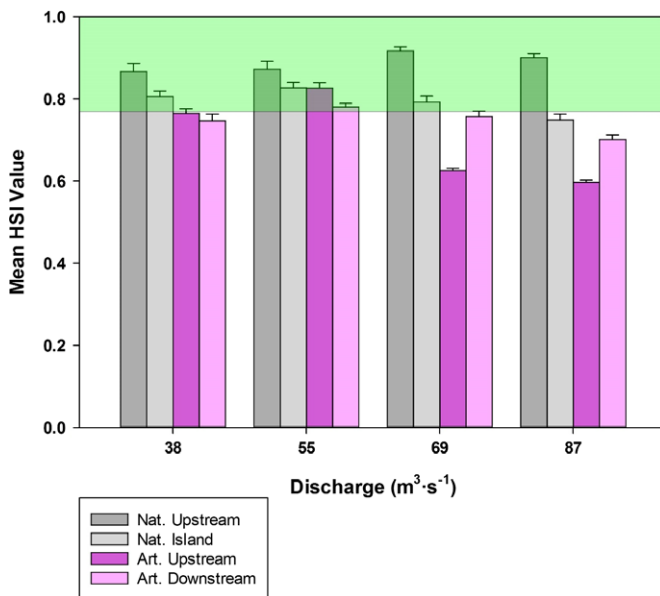


FIGURE 10. Mean habitat suitability index score for postrestoration (with weirs) conditions of documented Lake Sturgeon spawning sites on the Ouareau River for the four representative spawning discharges. The error bars represent the 95% CI. The green zone represents HSI values above the minimum suitability threshold. [Color figure can be viewed at afs-journals.org.]

mean of 8.8% (95% CI = 4.8%) of suitable habitat across all discharges (minimum = 2.6% at 87 m³/s, maximum = 13.7% at 38 m³/s). The natural island site had the most suitable area at discharges of 38 and 55 m³/s, whereas the natural upstream site had more suitable areas for discharges of 69 and 87 m³/s. There was no suitable habitat within the artificial upstream site for the two higher discharges.

In terms of habitat quality for postrestoration conditions, the natural upstream site had the highest mean HSI value for all discharges, followed by the natural island site (Figure 10). Notably, the mean conditions at the artificial sites were only considered suitable at the 55-m³/s discharge, while the natural sites were suitable for all discharges with the exception of the natural island site at 87 m³/s.

The dam activity upstream from the Ouareau River site raised concerns about flows being insufficient for spawning Lake Sturgeon, which is partly why the weirs were constructed. The postproject reports at the site indicated that water levels were raised by 0.80 m upstream from weir 1 for all discharges, whereas they ranged between 0.35 m and 0.55 m for weir 2 (Beaulieu 2010). However, the longitudinal water surface profile as predicted by the Delft3D simulations did not differ significantly between conditions with and without the weirs at the lowest discharge of 38 m³/s (Figure 11), where the impact of the weirs on the water surface should be greatest. The maximum difference in water surface elevation between the weir and no-weir scenarios predicted by the 3D model along the longitudinal profile was less than 0.06 m (Figure 11).

DISCUSSION

Although adding boulder weirs that trap gravel can increase the density of spawning salmon and steelhead *Oncorhynchus mykiss* in bedrock channels (Roni et al. 2008), the Ouareau River restoration project revealed several misconceptions about the impact of such structures for spawning Lake Sturgeon. First, unlike salmonids, Lake Sturgeon have a limited capacity to leap over obstacles during a spawning migration, so the weirs were designed to not exceed 0.3 m in height. The limited height of the structures explains why the 3D hydrodynamic simulations with and without weirs were very similar for velocity and flow depth. The impact of the submerged weirs on water levels were also markedly overestimated at the design stage (Figure 11), as flow depths were computed based on a theoretical equation that did not apply under submerged conditions and were not verified with a numerical hydrodynamic model. As this restoration project was expensive (approximately US\$500,000), using numerical modelling to assess the role of the planned instream structures and to verify whether suitable spawning habitat was

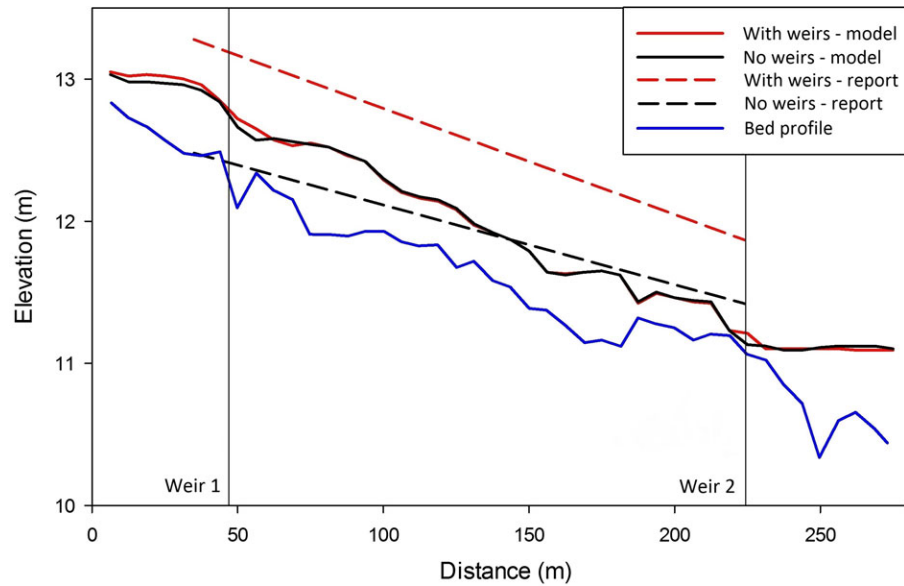


FIGURE 11. Ouareau River longitudinal profile for conditions with and without weirs at a flow of $38 \text{ m}^3/\text{s}$ for water surface elevation from the hydrodynamic model, the Beaulieu report (Beaulieu 2010), and bed elevation from the longitudinal bed profile (Beaulieu and Bélanger 2005). [Color figure can be viewed at afsjournals.org.]

available elsewhere in the reach would also have been a worthwhile investment. Indeed, when postproject evaluations for habitat restorations are conducted, studies typically focus on biological endpoints such as species abundance or richness (Ford 1989; Palmer et al. 2014). However, these projects often involve the manipulation of river form and function through the introduction of instream structures, which increase the cost of restoration projects, but are rarely assessed for performance (Bernhardt et al. 2005; O'Donnell and Galat 2007; Whiteway et al. 2010).

It is interesting to compare the Ouareau River project with the successful St. Clair–Detroit River system (SCDRS) project, which installed 24 spawning reefs at three sites in locations deprived of suitable habitat for Lake Sturgeon as well as Walleye *Sander vitreus*, Lake Whitefish *Coregonus clupeaformis*, and the Northern Madtom *Noturus stigmosus* (Manny et al. 2015; Krieger and Diana 2017). An adaptive management protocol (AMP) was used in the SCDRS (Figure 2 in Manny et al. 2015), revealing marked differences in restoration strategies compared with the Ouareau River case, which likely explain the different outcome of these two projects. The AMP process includes opportunities to adjust hypotheses or strategies once the physical and biological evaluations are completed (Manny et al. 2015). In the Ouareau River Lake Sturgeon project, both the physical and biological evaluations revealed that the project had not met its objectives, so the restoration strategies should have been revised.

The SCDRS restoration attempted to address anthropogenic disturbance as a result of dredging and

excavating coarse substrates (Manny et al. 2015). The project benefitted from a thorough assessment of the affected species in the system, the establishment of a clear hypothesis, the insights of a multidisciplinary committee, and the use of an iterative process with feedback and adjustment opportunities. As a result, the project identified areas that lacked adequate spawning habitat and created reefs that are now being used by the target species (Manny et al. 2015; Krieger and Diana 2017; Fischer et al. 2018). In contrast, the Ouareau River site was deemed to have experienced a loss of habitat quantity and function due to a landslide event in 1990 that covered the traditional spawning site (Figure 3) with clay and silt deposits. As landslides are very common in this area, as evidenced by several landslide scars in the Ouareau River valley (Bariteau and Duhamel 2008), it appears unlikely that such a natural disturbance would have greatly affected Lake Sturgeon spawning.

There is no clear evidence that there was a lack of adequate spawning habitat at the Ouareau River site. In a successful Lake Sturgeon restoration project on the Des Prairies River near Montreal, Dumont et al. (2011) cited a threshold for egg density of $3,000\text{--}3,500 \text{ eggs/m}^2$ to avoid high embryonic mortality rates, based on a study of Khoroshko and Vlasenko (1970) for Sevruga Sturgeon *A. stellatus*. Dumont et al. (2011) observed an increase in egg to larvae survival when egg densities decreased from a maximum of $5,626 \text{ eggs/m}^2$ (in 1996) to fewer than $2,000 \text{ eggs/m}^2$ (in 1997–1999). The project goal for the Ouareau River stated that egg densities should not exceed 500 eggs/m^2 . At the site, the maximum recorded egg

density before restoration was 963 eggs/m² (1999), and 645.7 eggs/m² after restoration (2007). Other than these two instances (out of 732 samples, or 0.27% of the time), there is no evidence of egg deposition exceeding the maximum threshold. The maximum mean density of eggs per spawning bed was 32.1 eggs/m² before restoration (2004) and 66.8 eggs/m² after restoration (2009). Given the low egg densities relative to the threshold at which a higher embryonic mortality rate is likely to occur based on the literature (Khoroshko and Vlasenko 1970; Dumont et al. 2011), it is unclear why a lack of adequate spawning habitat was stated as a concern at the Ouareau River site.

The lack of clarity regarding the purpose of the weirs points to a problem in scientific pre-assessment, consensus building, and communication, as the structures were initially stated as a means to resist headward erosion (Beaulieu and Bélanger 2005) and later as a means to maintain water levels (Chagnon and Dumas 2007; Beaulieu 2010). Our evaluation of the decision-making process showed that the necessity of weirs was conceived with no clear evidence of headward erosion, and that these structures were not designed to create a relevant backwater effect (Beaulieu and Bélanger 2005; Sear et al. 2010). Indeed, our study confirmed that they had a negligible effect in this regard. An expert committee consisting of biologists, engineers, project coordinators, and a technician were responsible for the conceptualization, planning, and execution of the restoration project. Expertise in hydrogeomorphology on the restoration committee would have likely identified these problems, but the fact that weirs have never been used in previous Lake Sturgeon restoration projects should also have raised concerns from biologists.

Because the restoration project on the Ouareau River did assess conditions at the site before and after restoration, we have the opportunity for a thorough evaluation of the project through our modelling approach and to offer new knowledge as part of the evaluation step of AMP. It seems clear that no more resources should be invested into maintaining the weirs, which required most of the budget allocated to this project. These structures have never been successfully installed on rivers that are larger than 40 m wide and experience discharges larger than 300 m³/s or velocities over 3 m/s (Sear et al. 2010), all of which are exceeded at the Ouareau River site.

The instream structures were designed to withstand a 100-year flood, estimated at 332.5 m³/s. However, this estimate did not take into account the presence of a large tributary, the Rouge River, between the gauging station and the study site (Figure 1), which resulted in an underestimation of discharge by a factor of 1.26. In reality, the “100-year flood” of 332.5 m³/s has a recurrence interval of 6 years. Since the installation of the weirs in 2007, this flood threshold has been exceeded three times, possibly leading to the degradation of the second weir.

Most importantly, these weirs did not provide any benefits to the spawning area and are therefore best left to degrade. For the artificial spawning reefs at the site, there is no clear need to restore them or create more. They were installed in locations that can enhance fine sediment infiltration into the reefs during low flow conditions, a critical factor for Lake Sturgeon spawning habitat (Johnson et al. 2006; Manny et al. 2015), and can experience average velocities above the suitable range for Lake Sturgeon during spawning flow conditions. This could have been verified with a 2D or 3D model before the installation of the structures at a much lower cost. It was also not clear that the creation of artificial spawning sites was needed since there are extensive areas with suitable habitat on the Ouareau River (Figures 7, 8).

The use of heavy machinery and hard engineering solutions to reconstruct channel form is often the first step to address concerns with fish habitat, in spite of a growing consensus to shift away from these methods as an initial response to perturbations (Palmer et al. 2014). When human intervention is necessary to create or enhance habitat, it is critical that thorough management protocols are adhered to and allow for multidisciplinary and multistakeholder input and review (Kondolf and Micheli 1995; Manny et al. 2015). Valuable insight can be gained through the comparison of projects that did and did not adopt an adaptive management protocol.

Management Implications

Failed restoration projects are seldom reviewed in the scientific literature despite the useful insights they can provide (Kondolf 1998; Bernhardt et al. 2005). This detailed assessment of the unsuccessful Ouareau River restoration project highlights the necessity in the future to work with an AMP similar to that of the successful St. Clair–Detroit River Lake Sturgeon restoration project (Manny et al. 2015). Part of the AMP should require multidisciplinary insights into aquatic habitat issues with a team consisting of river engineers, biologists, and hydrogeomorphologists to reduce the risk of unexpected problems with instream structure design such as the rapidly degrading boulder weir structures in the Ouareau River. Additionally, using a 2D or 3D numerical modelling approach before the implementation stages could greatly help reduce the cost of a restoration project and serve to adjust hypotheses (step 5 in the AMP of Manny et al. 2015: prioritization and strategy development), such as those concerning the impact of the weirs on water levels in the Ouareau River project. A one-dimensional hydrodynamic model (HEC-RAS) was used by the engineers in charge of the construction design in 2005. This precluded an analysis of the suitable habitat linked to spatial variability of depth and velocity, which can only be obtained in a 2D or 3D model. A 2D or 3D modelling approach would also have helped identify the

most appropriate location for artificial spawning sites. But, above all, this unsuccessful project highlights the importance of the very first step of AMP, namely identification of the problem. This might have revealed at the onset that this restoration project was unnecessary given the sufficient quantity of existing suitable spawning habitat in the Ouareau River.

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